



Hirusha Fernando

AI Engineer

Contact

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- 🌐 Portfolio Website

Skills

Programming

- Python
- JavaScript
- TypeScript
- C#

Frontend

- Angular
- Blazor
- Next.js

Backend

- FastAPI
- Django
- ASP.NET Core
- Entity Framework Core

Database

- MySQL
- MS SQL Server
- MongoDB
- Postgres
- PGVector
- RedisVector

AI/ML

- LangChain
- LangGraph
- Pandas
- scikit-learn
- CrewAI
- Hugging Face

Profile

Final year AI undergraduate and a passionate developer with 4+ years of experience developing web applications using Angular, Python, ASP.NET Core, EF Core, MS SQL, and Firebase. I am also interested in GenAI, AI Agents, Prompt Engineering and ML and familiar with vector databases, frameworks like LangChain, LangGraph. I am eager to be challenged to grow and further improve my skills.

Education

BSc (Hons) in Artificial Intelligence

2022–2026

University of Moratuwa

CGPA – 3.4 out of 4.00

G.C.E. Advanced Level

2017–2020

Maliyadeva College

Z Score – 2.1484

Experience

Associate Software Engineer – AI

2025 Aug – 2025 Oct

Ariso Technologies | Colombo

- Developed MCP tools and used them with LangGraph AI agents.
- Experienced with optimizing AI agents for reduce the response latency.
- Structuring AI agent outputs for WhatsApp automation.

Intern AI Engineer

2025 Feb – 2025 Aug

Ariso Technologies | Colombo

- Developed Agent Development Kit to manage AI agents with Python
- Worked with vector databases like PGVector, Qdrant, Redis Vector for implement memory for AI agents.
- Experience with HashiCorp Consul as a service discovery for AI Agents.
- Deployed and maintained AI agents on Linux servers.

Software Developer

2022 Jan – 2024 Dec

Crede Technologies | Canada (Remote)

- Hands-on experience developing web applications for the healthcare sector using ASP.NET Core, Blazor, EF Core, Ant Design Blazor, & MS SQL.
- Developed reusable components with Ant Design Blazor & Bootstrap.

Freelance Angular Developer

2022 Jan – 2023 Dec

Self-employed | Remote

- Developed and maintained web apps using Angular and Firebase.
- Collaborated with clients to design and implement responsive, scalable, and high-performance frontend solutions.

Other

- Git
- JIRA
- Docker
- GitHub Actions
- Linux
- Firebase

Soft Skills

Team Working

Adaptivity

Quick Learning

Problem solving

Languages

Sinhala - Native

English - Professional

Interest

Prompt Engineering

LLM Security

Listening Music

PC Gaming

References

Dr. Thilini Piyatilake

Senior Lecturer
University of Moratuwa
thilinispi@uom.lk
+94 77 117 7578

Mr. Shalitha Senanayaka

Senior Software Engineer
Digital Team
shalithax@gmail.com
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Trainee Web Developer

2020 Dec - 2022 Jan

Crede Technologies | Canada

- Implemented Agile practices for project management, leveraging Jira for enhanced team coordination
- Managed version control and code repositories with Git and BitBucket.
- Led the development of the core health portal using Django, jQuery, Bootstrap, and PostgreSQL, ensuring robust and scalable patient data management.

Projects

Detecting Prompt Injection Attacks With SLMs

A prompt injection detection system based on quantized Small Language Models and fine-tuned hot-swappable LoRA adapters. 4-bit quantized Gemma3-1B-it model as the backbone SLM and applies QLoRA for memory-efficient fine-tuning of category-specific LoRA adapters. Adapter based approach used for modular detection of different prompt injection categories while maintaining low computational overhead.

Technologies:

- Unslowth, Python, Kaggle Notebooks, Gemma3-1B-it SLM, Hugging Face

Call Center Communication Analyzer

[GitHub](#)

AI-based system to analyze customer feedback call recordings and provide insightful analytics and data visualizations.

Contribution:

- Full stack Development, Sentiment Analysis, Containerizing Backend, Creating CI/CD, Pipelines

Technologies:

- FastAPI, Angular, MongoDB, Redis, Celery, GitHub Actions, Docker, LangChain, AWS (S3 Bucket, EC2, Amazon Comprehend)

Online Ledger Management System

A ledger management system for a local client to manage their lending sale business customers' monthly payments and generate reports.

Contribution:

- Frontend Development

Technologies :

- Angular, Firebase(Firestore, Cloud Functions, Authentication), GitHub Actions

Blog Writing Assistant

[Github](#)

Agent based AI assistant for blog post writing in Medium platform. Take an existing blog as an input and gives improved blog article. It can generate or improve conclusion and introduction paragraphs, check grammar and spellings, generate SEO title and description for the blog and add markdown formatting for the blog.

Technologies:

- CrewAI, Streamlit, ChatGPT-4o LLM

ZoteroPDF

[Github](#)

ZoteroPDF is a custom plugin developed to address the lack of bulk PDF export functionality in Zotero. This plugin enables one-click export of all PDF attachments within a selected collection

Technologies:

- Typescript

SQL Based Question Answering RAG Application

SQL database-based question and answering LLM application using Retrieval Augmented Generation.

Technologies:

- LangChain, Streamlit, ChatGPT-4o LLM, PostgreSQL PGVector